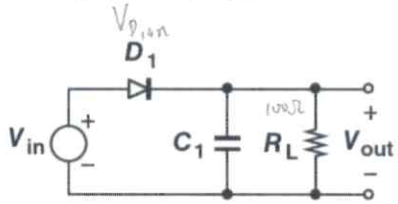
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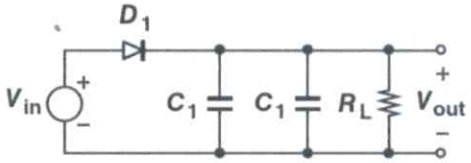
Midterm

5 Problems (20 points each, total 100 points)

1. The half-wave rectifier shown below drives a 100-Ω load through a diode with $V_{D,on} = 0.6\,\text{V}$ and a 500-μF smoothing capacitor as shown in (a). The input is a 60-Hz sinusoidal voltage source with a peak voltage of 3.5 V. If the circuit is replaced by (b), Calculate the ripple amplitude respectively (20 points)



(a)



(b)

(a) $V_R \approx \frac{I_L}{C_1 f_{in}}$, $I = \frac{V}{R}$

$V_{out,peak} = V_p - V_{D,on} = 3.5 - 0.6 = 2.9\,(V)$

$I_L = \frac{V_{out,peak}}{R_L} = \frac{2.9\,(V)}{100\,(\Omega)} = 29 \times 10^{-3}\,(A)$

$V_R \approx \frac{29 \times 10^{-3}}{(500 \times 10^{-6})(60)} = 0.967\,(V)$ #

(b) $C_{eq} = C_1 + C_1 = 1000 \times 10^{-6}$

$V_R \approx \frac{29 \times 10^{-3}}{(1000 \times 10^{-6})(60)} = 0.967 \times \frac{1}{2} = 0.483\,(V)$ #

A: (a) $V_R \approx 0.967\,(V)$

(b) $V_R \approx 0.483\,(V)$

2. A junction employs $N_D = 4 \times 10^{16}\ \text{cm}^{-3}$ and $N_A = 3 \times 10^{15}\text{cm}^{-3}$ Calculate the built-in potential at (a) T = 300 K and (b) 350 K. (20 points)

电位

$$V_0 = V_T \ln \frac{N_A N_D}{n_i^2}, \quad n_i = 5.2 \times 10^{15} \times T^{\frac{3}{2}} \exp\left(\frac{-Eg}{2kT}\right) \ (\text{electrons/cm}^3)$$

$$V_T = \frac{kT}{q}$$

$$Eg = 1.12\ (\text{eV}) = 1.792 \times 10^{-19}\ (\text{J})$$

$$k = 8.625 \times 10^{-5} \left(\frac{\text{eV}}{\text{K}}\right)$$

$$N_A N_D = (3 \times 10^{15})(4 \times 10^{16}) = 12 \times 10^{31}\ (\text{cm}^{-6})$$

$$\frac{k}{q} = 8.625 \times 10^{-5} \left(\frac{\text{V}}{\text{K}}\right)$$

(a) $T = 300\text{K}$

$$V_T = \frac{k}{q} T = (8.625 \times 10^{-5})(300) = 25.875\ (\text{mV}) = 25.875 \times 10^{-3}\ (\text{V})$$

$$n_i = 5.2 \times 10^{15} \times (300)^{\frac{3}{2}} \exp\left(\frac{-1.12}{2(8.625 \times 10^{-5})(300)}\right) = 1.0976 \times 10^{10}\ (\text{cm}^{-3})$$

$$V_0 = (25.875 \times 10^{-3}) \ln\left(\frac{25.875 \times 10^{-3}}{(1.0976 \times 10^{10})^2}\right) \approx 0.716\ (\text{V})\ \#$$

(b) $T = 350\text{K}$

$$V_T = \frac{k}{q} T = (8.625 \times 10^{-5})(350) = 30.1875\ (\text{mV}) = 30.1875 \times 10^{-3}\ (\text{V})$$

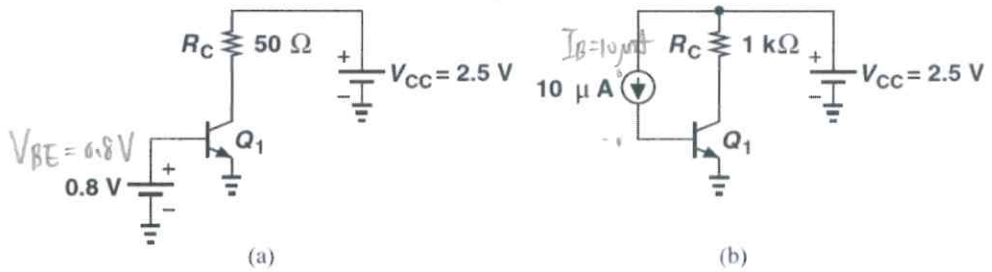
$$n_i = 5.2 \times 10^{15} \times (350)^{\frac{3}{2}} \exp\left(\frac{-1.12}{2(8.625 \times 10^{-5})(350)}\right) = 2.9891 \times 10^{11}\ (\text{cm}^{-3})$$

$$V_0 = (30.1875 \times 10^{-3}) \ln\left(\frac{30.1875 \times 10^{-3}}{(2.9891 \times 10^{11})^2}\right) \approx 0.634\ (\text{V})\ \#$$

A: (a) $V_0 \approx 0.716\ (\text{V})$

(b) $V_0 \approx 0.634\ (\text{V})$

3. Determine the operating point and the small-signal model of Q for the circuits shown in (a) and (b). Assume $I_S = 2\times10^{-16}$ A, $\beta = 100$, and $V_A = \infty$. (20points)



(a) ∵基极由0.8V电压源直接偏压,射极接地, ∴ $V_{BE} = 0.8$ V

$$I_C = I_S \exp\left(\frac{V_{BE}}{V_T}\right), \quad V_T = 26(mV) = 0.026(V)$$

$$I_C = (2\times10^{-16})\exp\left(\frac{0.8}{0.026}\right) = 4.61(mA)$$

$$I_B = \frac{I_C}{\beta} = \frac{4.61}{100} = 4.61\times10^{-2}(mA)$$

4.61
0.0461
≈ 4.661

$$I_E = I_C + I_B = 4.61 + 0.0461 = 4.6561 \approx 4.66(mA)$$

$$V_C = V_{CC} - I_C R_C = 2.5 - (4.61\,mA)(50\,\Omega) \approx 2.269(V)$$

∵ $V_C > V_B > V_E$ ∴基极-集极接反向偏,在主动区.

2.269 0.8 0

$$g_m = \frac{I_C}{V_T} = \frac{4.61(mA)}{26(mV)} \approx 177(mS), \quad r_\pi = \frac{\beta}{g_m} = \frac{100}{177(mS)} \approx 564(\Omega), \quad r_o = \frac{V_A}{I_C} = \infty$$

(b) ∵基极由10μA电流源驱动 ∴ $I_B = 10\mu A$

$$I_C = \beta I_B = (100)(10\mu A) = 1\,mA, \quad I_E = I_C + I_B = 1(mA) + 10(\mu A) = 1.01(mA)$$

$$V_C = V_{CC} - I_C R_C = 2.5 - (1\,mA)(1\,k\Omega) = 1.5(V)$$

$$V_{BE} = V_T \ln\frac{I_C}{I_S} = 26(mV)\ln\left(\frac{10^{-3}(A)}{2\times10^{-16}(A)}\right) \approx 760(mV)$$

∵ $V_C > V_{BE} > V_E$ ∴ 顺向主动区

1.5 0.76 0

$$g_m = \frac{I_C}{V_T} = \frac{1(mA)}{2.6(mV)} \approx 38.5\,mS, \quad r_\pi = \frac{\beta}{g_m} = \frac{100}{38.5(mS)} \approx 2.6(k\Omega), \quad r_o = \frac{V_A}{V_C} = \infty$$

A:

(a) $I_C = 4.61(mA)$, $I_B = 46.1(\mu A)$, $V_C \approx 2.269(V)$, $g_m \approx 177(mS)$, $r_\pi \approx 564(\Omega)$, $r_o = \infty$

(b) $I_C = 1(mA)$, $V_{BE} \approx 760(mV)$, $V_C = 1.5\,V$, $g_m \approx 38.5(mS)$, $r_\pi \approx 2.6(k\Omega)$, $r_o = \infty$

$I_C = 10^{-3}$ (A)

4. A bipolar transistor carries a collector current of 1mA with $V_{CE} = 2V$. Determine the required base-emitter voltage if (a) $V_A = \infty$ or (b) $V_A = 20V$. Assume $I_S = 3 \times 10^{-16}$ A.

求 V_{BE}

$$V_{BE} = V_T \ln\left(\frac{I_C}{I_S}\frac{1}{1+\frac{V_{CE}}{V_A}}\right) \quad \therefore V_T = 26(mV)$$

(a) $V_A = \infty$

$$V_{BE} = (26\,(mV)) \ln\left(\frac{10^{-3}}{3\times10^{-16}}\frac{1}{1+\frac{2}{\infty}}\right) = 750(mV) \#$$

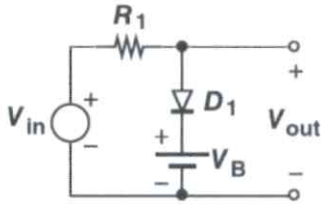
(b) $V_A = 20V$

$$V_{BE} = (26\,(mV)) \ln\left(\frac{10^{-3}}{3\times10^{-16}}\frac{1}{1+\frac{2}{20}}\right) = 747\,(mV) \#$$

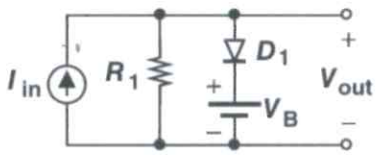
A: (a) $V_{BE} = 750\,(mV)$

(b) $V_{BE} = 747\,(mV)$

5. In the circuits of (a) and (b), plot the current flowing through R_1 as a function of V_{in} or I_{in}. Assume a constant-voltage diode model with $V_{Don} = 0.8V$ and $V_B = 1V$ (20points)



(a)



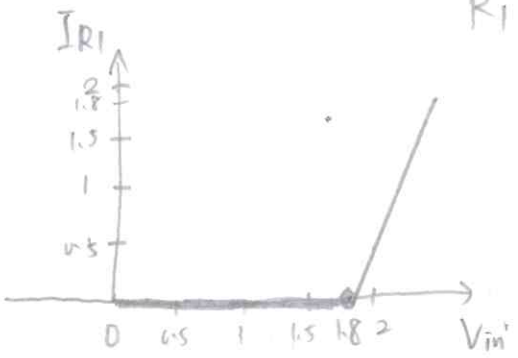
(b)

$V_{out} \geq V_B + V_{D,on} = 1(V) + 0.8(V) = 1.8\ (V)$

(a) 电压驱动

$V_{in} < 1.8V$: $I_{R_1} = 0$, $V_{out} = V_{in}$

D_1 导通 $V_{in} \geq 1.8V$: $I_{R_1} = \dfrac{V_{in} - V_{out}}{R_1} = \dfrac{V_{in} - 1.8}{R_1}$



(b) 电流驱动

$I_{in} < \dfrac{1.8V}{R_1}$: $I_{R_1} = I_{in}$

D_1 导通 $I_{in} \geq \dfrac{1.8V}{R_1}$: $I_{R_1} = \dfrac{1.8V}{R_1}$, $I_{D_1} = I_{in} - \dfrac{1.8V}{R_1}$

